

Complete Project Structure: RL Portfolio Budgeting System + Publications

Overview

This structure supports:

1. English Q1 journal article (LaTeX)
2. Persian [M.Sc.](#) thesis documentation (XePersian/LaTeX)
3. Full system implementation (RL agent, baselines, backend, dual frontend)
4. Reproducible experiments and data
5. Documentation and deployment guides
6. Strategic research planning and archival

Root Directory Structure

```
rl-portfolio-budgeting/
├─ README.md
├─ LICENSE
├─ .gitignore
├─ requirements.txt
├─
├─ strategy/
├─ literatureArchive/
├─ mscn/
├─
├─ paper/
│   ├─ main.tex
│   ├─ references.bib
│   ├─ sections/
│   │   ├─ 01-introduction.tex
│   │   ├─ 02-literature.tex
│   │   ├─ 03-problem-formulation.tex
│   │   ├─ 04-methodology.tex
│   │   ├─ 05-experiments.tex
│   │   ├─ 06-system-architecture.tex
│   │   ├─ 07-managerial-insights.tex
│   │   ├─ 08-conclusion.tex
│   │   └─ appendix.tex
│   ├─ figures/
│   │   ├─ architecture-diagram.pdf
│   │   ├─ speedup-comparison.pdf
│   │   ├─ quality-comparison.pdf
│   │   ├─ scalability-plot.pdf
│   │   ├─ sensitivity-analysis.pdf
│   │   ├─ attention-heatmap.pdf
│   │   └─ ui-screenshots/
│   │       ├─ web-interface.png
│   │       └─ desktop-interface.png
│   ├─ tables/
│   │   ├─ baseline-comparison.tex
│   │   ├─ scalability-results.tex
│   │   └─ finetuning-results.tex
│   └─ supplementary/
│       ├─ supplementary.tex
│       └─ hyperparameters.tex
```

```

| | | └─ additional-experiments.tex
| | | └─ code-snippets.tex
| | └─ submission/
| | | └─ manuscript.pdf
| | | └─ cover-letter.tex
| | | └─ response-to-reviewers.tex
| | | └─ highlights.txt
|
└─ thesis/
| | └─ main.tex
| | └─ references-fa.bib
| | └─ references-en.bib
| | └─ config/
| | | └─ preamble.tex
| | | └─ commands.tex
| | | └─ university-style.sty
| | └─ frontmatter/
| | | └─ title-page.tex
| | | └─ abstract-fa.tex
| | | └─ abstract-en.tex
| | | └─ acknowledgments.tex
| | | └─ dedication.tex
| | | └─ table-of-contents.tex
| | └─ chapters/
| | | └─ 01-introduction.tex # فصل اول: مقدمه
| | | └─ 02-literature.tex # فصل دوم: مروری بر ادبیات
| | | └─ 03-methodology.tex # فصل سوم: روش‌شناسی
| | | └─ 04-implementation.tex # فصل چهارم: پیاده‌سازی
| | | └─ 05-experiments.tex # فصل پنجم: آزمایش‌ها و نتایج
| | | └─ 06-system.tex # فصل ششم: سیستم نهایی
| | | └─ 07-conclusion.tex # فصل هفتم: نتیجه‌گیری
| | └─ appendices/
| | | └─ appendix-a-code.tex # پیوست الف: کدها
| | | └─ appendix-b-data.tex # پیوست ب: داده‌ها
| | | └─ appendix-c-results.tex # پیوست ج: نتایج تکمیلی
| | | └─ appendix-d-manual.tex # پیوست د: راهنمای استفاده
| | └─ figures/ # symlink -> ../paper/figures
| └─ submission/
| | └─ thesis-final.pdf
| | └─ defense-presentation.pdf
|
└─ src/
| └─ __init__.py

```

```
├─ config.py
├─ Makefile
├─ docker-compose.yml
├─ environment.yml
├─ environment/
│   ├── __init__.py
│   ├── generator.py
│   ├── distributions.py
│   ├── calibration.py
│   ├── validator.py
│   └─ portfolio.py
├─ agents/
│   ├── __init__.py
│   ├── ppo_agent.py
│   ├── network.py
│   ├── attention.py
│   ├── trainer.py
│   └─ finetuner.py
├─ baselines/
│   ├── __init__.py
│   ├── greedy.py
│   ├── mip_solver.py
│   ├── stochastic_prog.py
│   └─ perfect_info.py
├─ evaluation/
│   ├── __init__.py
│   ├── metrics.py
│   ├── comparator.py
│   ├── sensitivity.py
│   └─ interpretability.py
├─ backend/
│   ├── __init__.py
│   ├── app.py
│   ├── routes/
│   │   ├── __init__.py
│   │   ├── allocate.py
│   │   ├── finetune.py
│   │   ├── evaluate.py
│   │   └─ health.py
│   ├── models/
│   │   ├── __init__.py
│   │   ├── request.py
│   │   └─ response.py
```

```
|   └─ services/
|   |   └─ __init__.py
|   |   └─ inference.py
|   |   └─ finetuning.py
|   |   └─ storage.py
|   └─ utils/
|       └─ __init__.py
|       └─ validation.py
|       └─ logging.py
└─ frontend/
    └─ web/
        └─ app/
            └─ layout.tsx
            └─ page.tsx
            └─ upload/
                └─ page.tsx
            └─ allocate/
                └─ page.tsx
            └─ results/
                └─ page.tsx
            └─ finetune/
                └─ page.tsx
            └─ compare/
                └─ page.tsx
            └─ components/
                └─ ui/
                └─ ProjectTable.tsx
                └─ AllocationChart.tsx
                └─ MetricsCard.tsx
                └─ FileUpload.tsx
            └─ lib/
                └─ api.ts
                └─ utils.ts
                └─ types.ts
            └─ public/
            └─ styles/
                └─ globals.css
            └─ package.json
            └─ next.config.js
            └─ tsconfig.json
            └─ tailwind.config.js
    └─ desktop/
        └─ app/
```

```
|      |   ├── __init__.py
|      |   ├── main_window.py
|      |   ├── widgets/
|      |   |   ├── __init__.py
|      |   |   ├── project_table.py
|      |   |   ├── allocation_chart.py
|      |   |   ├── metrics_card.py
|      |   |   └── file_upload.py
|      |   ├── dialogs/
|      |   |   ├── __init__.py
|      |   |   ├── upload_dialog.py
|      |   |   ├── finetune_dialog.py
|      |   |   └── settings_dialog.py
|      |   ├── models/
|      |   |   ├── __init__.py
|      |   |   └── data_models.py
|      |   ├── services/
|      |   |   ├── __init__.py
|      |   |   └── api_client.py
|      |   └── assets/
|      |       ├── icons/
|      |       ├── images/
|      |       └── styles/
|      |           └── main.qss
|      ├── resources/
|      |   └── resources.qrc
|      ├── main.py
|      ├── requirements.txt
|      └── build.spec
└── experiments/
    ├── 01_environment_validation.py
    ├── 02_baseline_comparison.py
    ├── 03_rl_training.py
    ├── 04_scalability_analysis.py
    ├── 05_sensitivity_analysis.py
    ├── 06_finetuning_study.py
    ├── 07_interpretability.py
    └── 08_generate_figures.py
└── data/
    ├── synthetic/
    |   ├── parameters.json
    |   ├── instances/
    |   |   ├── train/
```

```
| | | | └─ val/
| | | |   └─ test/
| | |   └─ validation_stats.json
| | └─ finetuning/
| | | | └─ company_a/
| | | | └─ company_b/
| | |   └─ template.csv
| | └─ literature/
| | | | └─ cost_overrun_stats.csv
| | | | └─ delay_stats.csv
| | |   └─ risk_stats.csv
└─ models/
| | └─ pretrained/
| | | | └─ ppo_agent_final.pt
| | | | └─ ppo_agent_best.pt
| | |   └─ training_log.json
| | └─ finetuned/
| | | | └─ company_a_agent.pt
| | |   └─ company_b_agent.pt
| | └─ baselines/
| | |   └─ mip_solutions.pkl
└─ results/
| | └─ baseline_comparison/
| | | | └─ results.csv
| | | | └─ summary.json
| | |   └─ plots/
| | └─ scalability/
| | | | └─ timing_results.csv
| | |   └─ plots/
| | └─ sensitivity/
| | | | └─ uncertainty_sweep.csv
| | | | └─ budget_sweep.csv
| | |   └─ plots/
| | └─ finetuning/
| | | | └─ company_a_results.csv
| | | | └─ company_b_results.csv
| | |   └─ plots/
| | └─ interpretability/
| | | | └─ shap_values.csv
| | | | └─ attention_weights.csv
| | |   └─ plots/
└─ docs/
| | └─ README.md
```

- | |─ installation.md
- | |─ quickstart.md
- | |─ api-reference.md
- | |─ finetuning-guide.md
- | |─ deployment.md
- | |─ architecture.md
- | |─ experiments.md
- | |─ frontend-development.md
- | |─ contributing.md
- ─ tests/
 - | |─ __init__.py
 - | |─ test_environment.py
 - | |─ test_agents.py
 - | |─ test_baselines.py
 - | |─ test_backend.py
 - | |─ test_evaluation.py
 - | |─ test_frontend_api.py
- ─ scripts/
 - | |─ setup_environment.sh
 - | |─ download_dependencies.sh
 - | |─ generate_instances.py
 - | |─ train_agent.py
 - | |─ run_experiments.sh
 - | |─ generate_paper_figures.sh
 - | |─ build_desktop_app.sh
 - | |─ deploy_system.sh
- ─ docker/
 - | |─ Dockerfile.backend
 - | |─ Dockerfile.frontend-web
 - | |─ Dockerfile.experiments
 - | |─ nginx.conf
- ─ utils/
 - | |─ __init__.py
 - | |─ data_loader.py
 - | |─ visualization.py
 - | |─ logger.py
 - | |─ reproducibility.py

Key Design Decisions

1. Research Planning & Archival (`strategy/` , `literatureArchive/` , `mscn/`)

Centralized home for all strategic planning, decision rationale, and conversation history. The three sibling folders are yours to populate freely — kept separate from the codebase intentionally.

2. Dual Frontend Architecture

Next.js (`src/frontend/web/`) — build this first:

- Online deployment, reviewer demos, PhD committee access
- Next.js 14 App Router + TypeScript + Tailwind + shadcn/ui
- Deploy to Vercel or Docker + Nginx

PyQt (`src/frontend/desktop/`) — refactor from web after:

- Offline/enterprise use, distributable `.exe` / `.app`
- PyQt6 + PyInstaller for packaging
- Shares the same backend API — zero duplication of business logic

3. Shared Figures (Paper ↔ Thesis)

```
bash
```

```
cd thesis/figures
```

```
ln -s ../../paper/figures/* .
```

Generate once in `paper/figures/` , use in both documents.

4. Bilingual Bibliography

- `paper/references.bib` — English sources for journal
- `thesis/references-en.bib` — English sources with Persian annotations
- `thesis/references-fa.bib` — Persian sources

5. Reproducibility

Every result is script-generated from `src/experiments/` and saved to `src/results/` . No manual data manipulation anywhere.

LaTeX Templates

English Paper (paper/main.tex)

```
\documentclass[12pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath,amssymb,amsthm}
\usepackage{graphicx,booktabs}
\usepackage{algorithm,algorithmic}
\usepackage{hyperref,natbib}

\title{Reinforcement Learning for Real-Time Portfolio Budgeting Under Uncertainty:
      A Deployable Decision Support Framework}
\author{Your Name \ \ Department of Industrial Engineering \ \ University Name}

\begin{document}
\maketitle
\begin{abstract}% 150-200 words\end{abstract}

\section{Introduction} \input{sections/01-introduction}
\section{Literature Review} \input{sections/02-literature}
\section{Problem Formulation} \input{sections/03-problem-formulation}
\section{Methodology} \input{sections/04-methodology}
\section{Computational Experiments}\input{sections/05-experiments}
\section{System Architecture} \input{sections/06-system-architecture}
\section{Managerial Insights} \input{sections/07-managerial-insights}
\section{Conclusion} \input{sections/08-conclusion}

\bibliographystyle{elsarticle-harv}
\bibliography{references}
\appendix
\input{sections/appendix}
\end{document}
```

Persian Thesis (thesis/main.tex)

```
\documentclass[12pt,a4paper]{report}
\usepackage{xepersian}
\settextfont{XB Niloofar}
\setdigitfont{XB Niloofar}

\input{config/preamble}
\input{config/commands}

\begin{document}
\input{frontmatter/title-page}
\input{frontmatter/abstract-fa}
\input{frontmatter/abstract-en}
\input{frontmatter/acknowledgments}
\input{frontmatter/dedication}
\tableofcontents\listoffigures\listoftables

\input{chapters/01-introduction}
\input{chapters/02-literature}
\input{chapters/03-methodology}
\input{chapters/04-implementation}
\input{chapters/05-experiments}
\input{chapters/06-system}
\input{chapters/07-conclusion}

\appendix
\input{appendices/appendix-a-code}
\input{appendices/appendix-b-data}
\input{appendices/appendix-c-results}
\input{appendices/appendix-d-manual}

\bibliographystyle{ieeetr-fa}
\bibliography{references-fa,references-en}
\end{document}
```

src/Makefile

```
.PHONY: help install data train experiments figures test \  
        backend frontend-web frontend-desktop build-desktop \  
        deploy paper thesis clean
```

help:

```
@echo "install          Install all dependencies"  
@echo "data              Generate synthetic data"  
@echo "train              Train RL agent"  
@echo "experiments        Run all experiments"  
@echo "figures            Generate paper figures"  
@echo "test              Run unit tests"  
@echo "backend            Start FastAPI backend"  
@echo "frontend-web       Start Next.js dev server"  
@echo "frontend-desktop   Run PyQt desktop app"  
@echo "build-desktop      Build PyQt executable"  
@echo "deploy            Deploy via Docker"  
@echo "paper             Compile English paper (pdflatex)"  
@echo "thesis            Compile Persian thesis (xelatex)"  
@echo "clean             Remove generated files"
```

install:

```
pip install -r ../requirements.txt  
cd frontend/web && npm install  
cd frontend/desktop && pip install -r requirements.txt
```

data:

```
python scripts/generate_instances.py
```

train:

```
python scripts/train_agent.py
```

experiments:

```
bash scripts/run_experiments.sh
```

figures:

```
bash scripts/generate_paper_figures.sh
```

test:

```
pytest tests/
```

backend:

```
cd backend && uvicorn app:app --reload
```

frontend-web:

```
cd frontend/web && npm run dev
```

frontend-desktop:

```
cd frontend/desktop && python main.py
```

build-desktop:

```
bash scripts/build_desktop_app.sh
```

deploy:

```
docker-compose up -d
```

paper:

```
cd ../../paper && pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex ma
```

thesis:

```
cd ../../thesis && xelatex main.tex && bibtex main && xelatex main.tex && xelatex main
```

clean:

```
find . -name "*.pyc" -delete
```

```
find . -name "__pycache__" -type d -exec rm -rf {} +
```

```
cd ../../paper && rm -f *.aux *.log *.bbl *.blg *.out *.toc *.lof *.lot
```

```
cd ../../thesis && rm -f *.aux *.log *.bbl *.blg *.out *.toc *.lof *.lot
```

```
cd frontend/web && rm -rf .next
```

src/docker-compose.yml

```
services:
  backend:
    build:
      context: ..
      dockerfile: src/docker/Dockerfile.backend
    ports:
      - "8000:8000"
    volumes:
      - ./models:/app/models

  frontend-web:
    build:
      context: ./frontend/web
      dockerfile: ../../docker/Dockerfile.frontend-web
    ports:
      - "3000:3000"
    environment:
      - NEXT_PUBLIC_API_URL=http://backend:8000

  nginx:
    image: nginx:alpine
    ports:
      - "80:80"
    volumes:
      - ./docker/nginx.conf:/etc/nginx/nginx.conf
    depends_on:
      - backend
      - frontend-web
```

src/environment.yml

name: rl-portfolio

channels:

- defaults
- conda-forge

dependencies:

- python=3.11
- pip
- numpy
- pandas
- scipy
- matplotlib
- seaborn
- jupyter
- pip:
 - torch
 - stable-baselines3
 - gymnasium
 - fastapi
 - uvicorn
 - pydantic
 - pulp
 - pyomo
 - shap
 - pytest
 - black
 - ruff

.gitignore

```
# Python
__pycache__/
*.pyc
*.egg-info/
.pytest_cache/

# Large files (use Git LFS instead)
src/data/synthetic/instances/
src/models/pretrained/*.pt
src/models/finetuned/*.pt
src/results/

# LaTeX
*.aux *.log *.bbl *.blg *.out
*.toc *.lof *.lot *.synctex.gz

# Next.js
src/frontend/web/node_modules/
src/frontend/web/.next/
src/frontend/web/out/
src/frontend/web/.env*.local

# PyQt build
src/frontend/desktop/build/
src/frontend/desktop/dist/

# IDE / OS
.vscode/
.idea/
.DS_Store
Thumbs.db

### Git LFS
bash
git lfs track "*.pt"
git lfs track "*.pkl"
git lfs track "src/data/literature/*.csv"
```

Implementation Timeline

Week	Phase	Key Deliverables
1–3	Foundation	src/environment/ , src/data/synthetic/ , strategy/ setup
4–7	Baselines	src/baselines/ , src/results/baseline_comparison/
8–11	RL Agent	src/agents/ , src/models/pretrained/
12–16	System Development	src/backend/ , src/frontend/web/ , src/frontend/desktop/
17–19	Analysis	src/experiments/04-08 , src/results/ , paper/figures/
20–22	Writing	paper/sections/ , thesis/chapters/ , submission files

Critical Success Factors

1. Create all directories and placeholder files in Week 1
2. Commit after each experiment — not at the end
3. Generate figures once in paper/figures/ , symlink from thesis/
4. Every result must be reproducible from a script
5. Write src/docs/ as you build, not at the end
6. Build Next.js first, refactor to PyQt — share the backend API throughout